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HR Attrition Prediction and Retention Strategy Using Machine Learning

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1 Introduction

This project builds a KNN model to predict whether an employee will leave or stay with their employer.

The dataset includes features such as satisfaction, workload, salary, and department.

The model outputs: - 0 = stayed - 1 = left

2 Step 1: Define the Problem

We are predicting whether an employee will leave.

Target variable: - `left` (1 = left, 0 = stayed)

This is a binary classification problem.

3 Step 2: Get and Clean the Data

Load the dataset and prepare it for modeling.

```
hr <- read.csv("../data/hrdata.csv")
```

remove unnecessary columns

```
hr$X <- NULL
```

```
hr$dept <- hr$sales
hr$sales <- NULL
```

convert categorical variables to factors

```
hr$salary <- as.factor(hr$salary)
hr$dept <- as.factor(hr$dept)
```

verify data structure

```
str(hr)
```

```
## 'data.frame':    14999 obs. of  10 variables:
##  $ satisfaction_level   : num  0.38 0.8 0.11 0.72 0.37 0.41 0.1 0.92 0.89 0.42 ...
##  $ last_evaluation      : num  0.53 0.86 0.88 0.87 0.52 0.5 0.77 0.85 1 0.53 ...
##  $ number_project       : int  2 5 7 5 2 2 6 5 5 2 ...
##  $ average_montly_hours : int  157 262 272 223 159 153 247 259 224 142 ...
##  $ time_spend_company   : int  3 6 4 5 3 3 4 5 5 3 ...
##  $ Work_accident        : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ left                 : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ promotion_last_5years: int  0 0 0 0 0 0 0 0 0 0 ...
##  $ salary               : Factor w/ 3 levels "high","low","medium": 2 3 3 2 2 2 2 2 2 2 ...
##  $ dept                 : Factor w/ 10 levels "accounting","hr",..: 8 8 8 8 8 8 8 8 8 8 ...
```

View summary statistics

```
summary(hr)
```

```
##  satisfaction_level last_evaluation  number_project  average_montly_hours
##  Min.   :0.0900     Min.   :0.3600   Min.   :2.000   Min.   : 96.0
##  1st Qu.:0.4400     1st Qu.:0.5600   1st Qu.:3.000   1st Qu.:156.0
##  Median :0.6400     Median :0.7200   Median :4.000   Median :200.0
##  Mean   :0.6128     Mean   :0.7161   Mean   :3.803   Mean   :201.1
##  3rd Qu.:0.8200     3rd Qu.:0.8700   3rd Qu.:5.000   3rd Qu.:245.0
##  Max.   :1.0000     Max.   :1.0000   Max.   :7.000   Max.   :310.0
##
##  time_spend_company Work_accident        left        promotion_last_5years
##  Min.   : 2.000     Min.   :0.0000   Min.   :0.0000   Min.   :0.00000
##  1st Qu.: 3.000     1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.00000
##  Median : 3.000     Median :0.0000   Median :0.0000   Median :0.00000
##  Mean   : 3.498     Mean   :0.1446   Mean   :0.2381   Mean   :0.02127
##  3rd Qu.: 4.000     3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.00000
##  Max.   :10.000     Max.   :1.0000   Max.   :1.0000   Max.   :1.00000
##
##    salary             dept
##  high  :1237   sales      :4140
##  low   :7316   technical  :2720
##  medium:6446   support    :2229
##                IT         :1227
##                product_mng: 902
##                marketing  : 858
##                (Other)    :2923
```

```
colSums(is.na(hr))
```

```
##    satisfaction_level       last_evaluation        number_project
##                     0                     0                     0
##  average_montly_hours    time_spend_company         Work_accident
##                     0                     0                     0
##                  left promotion_last_5years                salary
##                     0                     0                     0
##                  dept
##                     0
```

```
summary(hr)
```

```
##  satisfaction_level last_evaluation  number_project  average_montly_hours
##  Min.   :0.0900     Min.   :0.3600   Min.   :2.000   Min.   : 96.0
##  1st Qu.:0.4400     1st Qu.:0.5600   1st Qu.:3.000   1st Qu.:156.0
##  Median :0.6400     Median :0.7200   Median :4.000   Median :200.0
##  Mean   :0.6128     Mean   :0.7161   Mean   :3.803   Mean   :201.1
##  3rd Qu.:0.8200     3rd Qu.:0.8700   3rd Qu.:5.000   3rd Qu.:245.0
##  Max.   :1.0000     Max.   :1.0000   Max.   :7.000   Max.   :310.0
##
##  time_spend_company Work_accident        left        promotion_last_5years
##  Min.   : 2.000     Min.   :0.0000   Min.   :0.0000   Min.   :0.00000
##  1st Qu.: 3.000     1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.00000
##  Median : 3.000     Median :0.0000   Median :0.0000   Median :0.00000
##  Mean   : 3.498     Mean   :0.1446   Mean   :0.2381   Mean   :0.02127
##  3rd Qu.: 4.000     3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.00000
##  Max.   :10.000     Max.   :1.0000   Max.   :1.0000   Max.   :1.00000
##
##    salary              dept
##  high  :1237   sales      :4140
##  low   :7316   technical  :2720
##  medium:6446   support    :2229
##                IT         :1227
##                product_mng: 902
##                marketing  : 858
##                (Other)    :2923
```

Step 3: Encode categorical variables

```
#model.matrix() creates dummy variables for factors
hr_dummies <- as.data.frame(model.matrix(~ . -1, data = hr))

# View the structure
str(hr_dummies)
```

```
## 'data.frame':    14999 obs. of  20 variables:
##  $ satisfaction_level   : num  0.38 0.8 0.11 0.72 0.37 0.41 0.1 0.92 0.89 0.42 ...
##  $ last_evaluation      : num  0.53 0.86 0.88 0.87 0.52 0.5 0.77 0.85 1 0.53 ...
##  $ number_project       : num  2 5 7 5 2 2 6 5 5 2 ...
##  $ average_montly_hours : num  157 262 272 223 159 153 247 259 224 142 ...
##  $ time_spend_company   : num  3 6 4 5 3 3 4 5 5 3 ...
##  $ Work_accident        : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ left                 : num  1 1 1 1 1 1 1 1 1 1 ...
##  $ promotion_last_5years: num  0 0 0 0 0 0 0 0 0 0 ...
##  $ salaryhigh           : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ salarylow            : num  1 0 0 1 1 1 1 1 1 1 ...
##  $ salarymedium         : num  0 1 1 0 0 0 0 0 0 0 ...
##  $ depthr               : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ deptIT               : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ deptmanagement       : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ deptmarketing        : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ deptproduct_mng      : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ deptRandD            : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ deptsales            : num  1 1 1 1 1 1 1 1 1 1 ...
##  $ deptsupport          : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ depttechnical        : num  0 0 0 0 0 0 0 0 0 0 ...
```

Normalize all numeric values to range 0–1

```
#Min-Max scaling function
minmax <- function(x) {
  (x - min(x)) / (max(x) - min(x))
}

#Apply to all columns
hr_norm <- as.data.frame(lapply(hr_dummies, minmax))

# View summary to confirm values are between 0 and 1
summary(hr_norm)
```

```
## satisfaction_level last_evaluation  number_project   average_montly_hours
## Min.   :0.0000     Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
## 1st Qu.:0.3846     1st Qu.:0.3125   1st Qu.:0.2000   1st Qu.:0.2804
## Median :0.6044     Median :0.5625   Median :0.4000   Median :0.4860
## Mean   :0.5745     Mean   :0.5564   Mean   :0.3606   Mean   :0.4909
## 3rd Qu.:0.8022     3rd Qu.:0.7969   3rd Qu.:0.6000   3rd Qu.:0.6963
## Max.   :1.0000     Max.   :1.0000   Max.   :1.0000   Max.   :1.0000
## time_spend_company Work_accident        left        promotion_last_5years
## Min.   :0.0000     Min.   :0.0000   Min.   :0.0000   Min.   :0.00000
## 1st Qu.:0.1250     1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.00000
## Median :0.1250     Median :0.0000   Median :0.0000   Median :0.00000
## Mean   :0.1873     Mean   :0.1446   Mean   :0.2381   Mean   :0.02127
## 3rd Qu.:0.2500     3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.00000
## Max.   :1.0000     Max.   :1.0000   Max.   :1.0000   Max.   :1.00000
##    salaryhigh        salarylow        salarymedium        depthr
## Min.   :0.00000   Min.   :0.0000   Min.   :0.0000   Min.   :0.00000
## 1st Qu.:0.00000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.00000
## Median :0.00000   Median :0.0000   Median :0.0000   Median :0.00000
## Mean   :0.08247   Mean   :0.4878   Mean   :0.4298   Mean   :0.04927
## 3rd Qu.:0.00000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:0.00000
## Max.   :1.00000   Max.   :1.0000   Max.   :1.0000   Max.   :1.00000
##     deptIT        deptmanagement  deptmarketing    deptproduct_mng
## Min.   :0.00000   Min.   :0.000   Min.   :0.0000   Min.   :0.00000
## 1st Qu.:0.00000   1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.00000
## Median :0.00000   Median :0.000   Median :0.0000   Median :0.00000
## Mean   :0.08181   Mean   :0.042   Mean   :0.0572   Mean   :0.06014
## 3rd Qu.:0.00000   3rd Qu.:0.000   3rd Qu.:0.0000   3rd Qu.:0.00000
## Max.   :1.00000   Max.   :1.000   Max.   :1.0000   Max.   :1.00000
##    deptRandD         deptsales       deptsupport     depttechnical
## Min.   :0.00000   Min.   :0.000   Min.   :0.0000   Min.   :0.0000
## 1st Qu.:0.00000   1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.0000
## Median :0.00000   Median :0.000   Median :0.0000   Median :0.0000
## Mean   :0.05247   Mean   :0.276   Mean   :0.1486   Mean   :0.1813
## 3rd Qu.:0.00000   3rd Qu.:1.000   3rd Qu.:0.0000   3rd Qu.:0.0000
## Max.   :1.00000   Max.   :1.000   Max.   :1.0000   Max.   :1.0000
```

Step 4: Split the Data

```
set.seed(12345)  # for reproducibility (so you get the same split each time)
```

```
#Pick 30% of rows randomly for testing
test_rows <- sample(1:nrow(hr_norm), 0.3 * nrow(hr_norm))

#Split into training and test sets
hr_test <- hr_norm[test_rows, ]
hr_train <- hr_norm[-test_rows, ]

#Save the target variable (left) separately for both sets
hr_test_y <- hr_test$left
hr_train_y <- hr_train$left

#Remove 'left' column from features (because we already saved it)
hr_test$left <- NULL
hr_train$left <- NULL
```

```
#check data
nrow(hr_train)  # should be about 70% of the total
```

```
## [1] 10500
```

```
nrow(hr_test)   # should be about 30% of the total
```

```
## [1] 4499
```

```
head(hr_train_y)  # target variable
```

```
## [1] 1 1 1 1 1 1
```

```
head(hr_train)    # feature columns
```

```
##   satisfaction_level last_evaluation number_project average_montly_hours
## 1        0.31868132       0.265625             0.0          0.2850467
## 3        0.02197802       0.812500             1.0          0.8224299
## 4        0.69230769       0.796875             0.6          0.5934579
## 5        0.30769231       0.250000             0.0          0.2943925
## 6        0.35164835       0.218750             0.0          0.2663551
## 7        0.01098901       0.640625             0.8          0.7056075
##   time_spend_company Work_accident promotion_last_5years salaryhigh salarylow
## 1              0.125            0                    0          0         1
## 3              0.250            0                    0          0         0
## 4              0.375            0                    0          0         1
## 5              0.125            0                    0          0         1
## 6              0.125            0                    0          0         1
## 7              0.250            0                    0          0         1
##   salarymedium depthr deptIT deptmanagement deptmarketing deptproduct_mng
## 1            0     0      0             0             0               0
## 3            1     0      0             0             0               0
## 4            0     0      0             0             0               0
## 5            0     0      0             0             0               0
## 6            0     0      0             0             0               0
## 7            0     0      0             0             0               0
##   deptRandD deptsales deptsupport depttechnical
## 1         0         1           0             0
## 3         0         1           0             0
## 4         0         1           0             0
## 5         0         1           0             0
## 6         0         1           0             0
## 7         0         1           0             0
```

Step 5: Build the Model and Predict ## LR

```
# --- Prepare data (attach target properly) ---
hr_train_lr <- hr_train
hr_train_lr$Attrition <- hr_train_y

hr_test_lr <- hr_test
hr_test_lr$Attrition <- hr_test_y

# --- Train Logistic Regression ---
lr_model <- glm(Attrition ~ ., data = hr_train_lr, family = "binomial")

# --- Predict probabilities ---
lr_probs <- predict(lr_model, newdata = hr_test_lr, type = "response")

# --- Convert probabilities to class labels (threshold = 0.3 for better recall) ---
lr_pred <- ifelse(lr_probs > 0.3, 1, 0)

# --- Confusion Matrix ---
conf_matrix <- table(Predicted = lr_pred, Actual = hr_test_lr$Attrition)
knitr::kable(conf_matrix, caption = "Confusion Matrix")
```

Confusion Matrix

	0	1
0	2787	350
1	618	744

```
# --- Evaluation Metrics ---
accuracy <- mean(lr_pred == hr_test_lr$Attrition)

sensitivity <- sum(lr_pred == 1 & hr_test_lr$Attrition == 1) /
               sum(hr_test_lr$Attrition == 1)

precision <- sum(lr_pred == 1 & hr_test_lr$Attrition == 1) /
             sum(lr_pred == 1)

metrics <- data.frame(
  Metric = c("Accuracy", "Sensitivity", "Precision"),
  Value = c(accuracy, sensitivity, precision)
)

knitr::kable(metrics, digits = 3, caption = "Model Performance Metrics")
```

Model Performance Metrics

Metric	Value
Accuracy	0.785
Sensitivity	0.680
Precision	0.546

```
# --- Logistic Regression Coefficients ---
lr_coefficients <- summary(lr_model)$coefficients

knitr::kable(
  round(lr_coefficients, 3),
  caption = "Logistic Regression Coefficients"
)
```

Logistic Regression Coefficients

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	0.155	0.139	1.114	0.265
satisfaction_level	-3.719	0.107	-34.826	0.000
last_evaluation	0.443	0.114	3.886	0.000
number_project	-1.490	0.127	-11.760	0.000
average_montly_hours	0.893	0.133	6.731	0.000
time_spend_company	2.076	0.148	14.031	0.000
Work_accident	-1.676	0.112	-14.954	0.000
promotion_last_5years	-1.358	0.291	-4.660	0.000
salaryhigh	-1.282	0.148	-8.679	0.000
salarylow	0.532	0.055	9.723	0.000
depthr	0.116	0.157	0.737	0.461
deptIT	-0.203	0.145	-1.401	0.161
deptmanagement	-0.478	0.192	-2.489	0.013
deptmarketing	-0.079	0.159	-0.496	0.620

	Estimate	Std. Error	z value	Pr(>|z|)
deptproduct_mng	-0.061	0.154	-0.393	0.695
deptRandD	-0.604	0.175	-3.458	0.001
deptsales	-0.019	0.121	-0.155	0.877
deptsupport	0.042	0.129	0.325	0.746
depttechnical	0.073	0.126	0.583	0.560

```
# --- Odds Ratios (VERY important for interpretation) ---
odds_ratios <- exp(coef(lr_model))

odds_table <- data.frame(
  Feature = names(odds_ratios),
  Odds_Ratio = round(odds_ratios, 3)
)

knitr::kable(odds_table, caption = "Odds Ratios (Interpretation)")
```

Odds Ratios (Interpretation)

	Feature	Odds_Ratio
(Intercept)	(Intercept)	1.167
satisfaction_level	satisfaction_level	0.024
last_evaluation	last_evaluation	1.557
number_project	number_project	0.225
average_montly_hours	average_montly_hours	2.441
time_spend_company	time_spend_company	7.969
Work_accident	Work_accident	0.187
promotion_last_5years	promotion_last_5years	0.257
salaryhigh	salaryhigh	0.278
salarylow	salarylow	1.702
salarymedium	salarymedium	NA
depthr	depthr	1.123

	Feature	Odds_Ratio
deptIT	deptIT	0.816
deptmanagement	deptmanagement	0.620
deptmarketing	deptmarketing	0.924
deptproduct_mng	deptproduct_mng	0.941
deptRandD	deptRandD	0.547
deptsales	deptsales	0.981
deptsupport	deptsupport	1.043
depttechnical	depttechnical	1.076

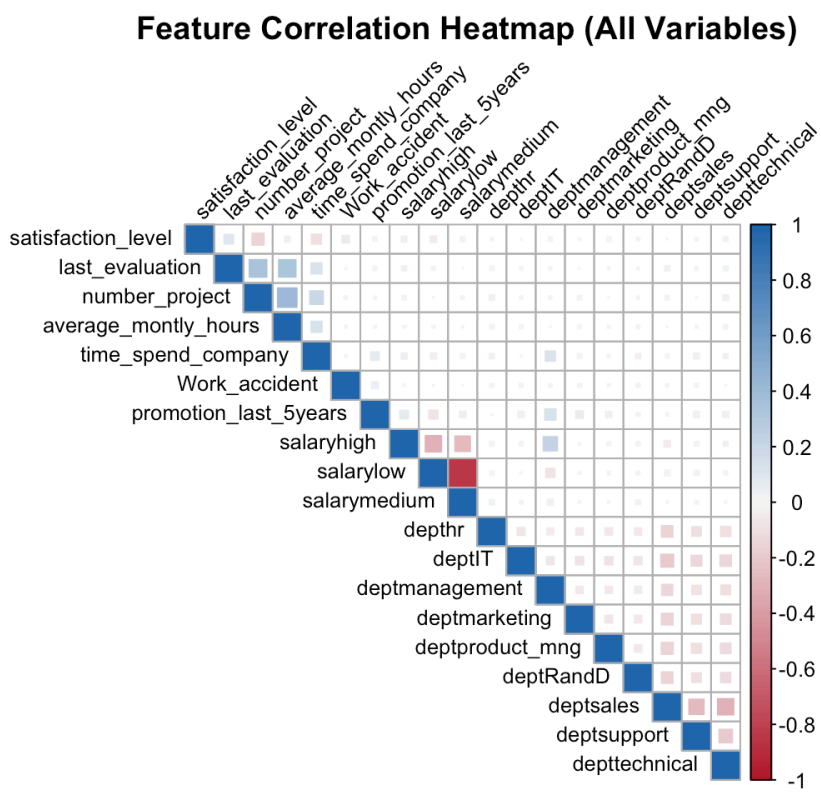
```
# --- Clean Feature Importance Table ---
lr_feature_table <- data.frame(
  Feature = names(coef(lr_model)),
  Coefficient = coef(lr_model),
  Odds_Ratio = exp(coef(lr_model))
)

# Sort by impact
lr_feature_table <- lr_feature_table[
  order(abs(lr_feature_table$Coefficient), decreasing = TRUE),
]

knitr::kable(
  lr_feature_table,
  digits = 3,
  caption = "Feature Importance (Sorted by Impact)"
)
```

Feature Importance (Sorted by Impact)

	Feature	Coefficient	Odds_Ratio
satisfaction_level	satisfaction_level	-3.719	0.024
time_spend_company	time_spend_company	2.076	7.969
Work_accident	Work_accident	-1.676	0.187
number_project	number_project	-1.490	0.225

	Feature	Coefficient	Odds_Ratio
promotion_last_5years	promotion_last_5years	-1.358	0.257
salaryhigh	salaryhigh	-1.282	0.278
average_montly_hours	average_montly_hours	0.893	2.441
deptRandD	deptRandD	-0.604	0.547
salarylow	salarylow	0.532	1.702
deptmanagement	deptmanagement	-0.478	0.620
last_evaluation	last_evaluation	0.443	1.557
deptIT	deptIT	-0.203	0.816
(Intercept)	(Intercept)	0.155	1.167
depthr	depthr	0.116	1.123
deptmarketing	deptmarketing	-0.079	0.924
depttechnical	depttechnical	0.073	1.076
deptproduct_mng	deptproduct_mng	-0.061	0.941
deptsupport	deptsupport	0.042	1.043
deptsales	deptsales	-0.019	0.981
salarymedium	salarymedium	NA	NA

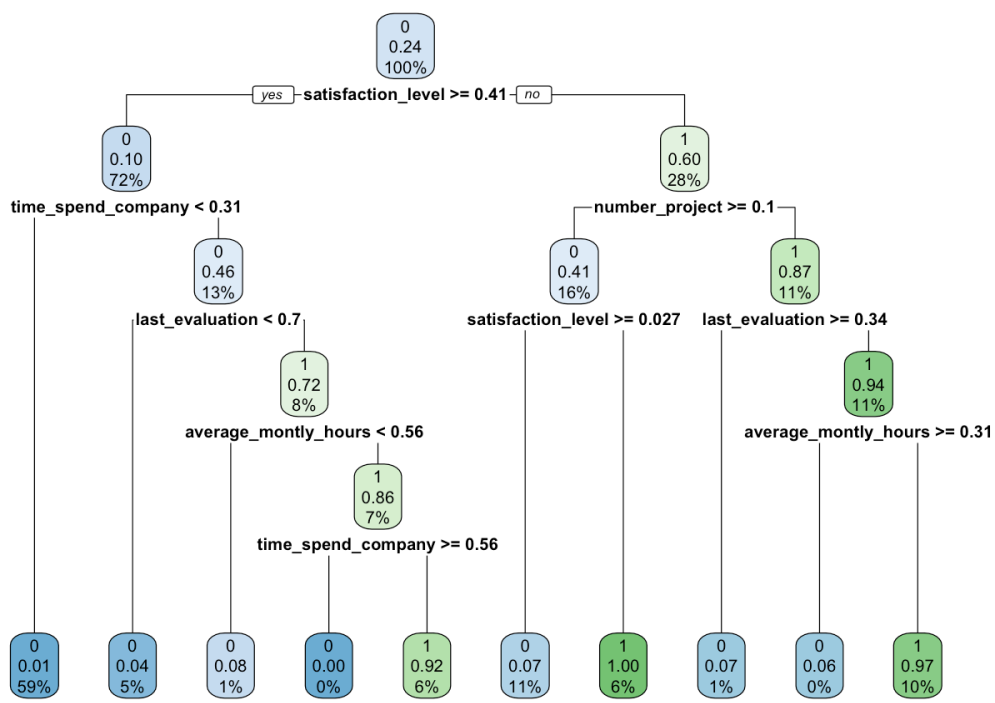
```
# Select numeric features
numeric_data <- hr_train[, sapply(hr_train, is.numeric)]

# Correlation matrix
cor_matrix <- cor(numeric_data, use = "complete.obs")

corrplot(
  cor_matrix,
  method = "square",
  type = "upper",
  diag = TRUE,
  tl.col = "black",
  tl.cex = 0.8,
  tl.srt = 45,
  addCoef.col = NULL,   # remove numbers
  addgrid.col = "grey70",
  cl.cex = 0.8,
  mar = c(0, 0, 2, 0),
  col = colorRampPalette(c("#B2182B", "#F7F7F7", "#2166AC"))(200),
  title = "Feature Correlation Heatmap (All Variables)"
)
```



3.1 DT

```
# Create training and test data frames with target variable
hr_train_df <- data.frame(hr_train)
hr_train_df$left <- as.factor(hr_train_y)

hr_test_df <- data.frame(hr_test)
hr_test_df$left <- as.factor(hr_test_y)

# Train decision tree model
tree_model <- rpart(
  left ~ .,
  data = hr_train_df,
  method = "class",
  control = rpart.control(cp = 0.01)
)
```

```
rpart.plot(tree_model)
```



```
# Predict probabilities on test data
tree_probs <- predict(tree_model, newdata = hr_test_df, type = "prob")

# Probability of leaving = column 2
tree_probs <- tree_probs[, 2]

# Convert probabilities to class labels using 0.5 threshold
tree_pred <- ifelse(tree_probs > 0.5, 1, 0)
```

```
library(knitr)
library(kableExtra)

# Confusion matrix
conf_matrix_tree <- table(Predicted = tree_pred, Actual = hr_test_y)

conf_df <- as.data.frame.matrix(conf_matrix_tree)
colnames(conf_df) <- c("Actual: 0 (Stayed)", "Actual: 1 (Left)")
conf_df <- cbind(Predicted = rownames(conf_df), conf_df)
rownames(conf_df) <- NULL

kable(conf_df, caption = "Decision Tree Confusion Matrix") %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover", "conden
sed"))
```

Decision Tree Confusion Matrix

Predicted	Actual: 0 (Stayed)	Actual: 1 (Left)
0	3364	92
1	41	1002

```
# Metrics
tree_accuracy <- mean(tree_pred == hr_test_y)
tree_sensitivity <- sum(tree_pred == 1 & hr_test_y == 1) / sum(hr_test_y == 1)
tree_precision <- sum(tree_pred == 1 & hr_test_y == 1) / sum(tree_pred == 1)

metrics_df <- data.frame(
  Metric = c("Accuracy", "Recall (Sensitivity)", "Precision"),
  Value = round(c(tree_accuracy, tree_sensitivity, tree_precision), 3)
)

kable(metrics_df, caption = "Decision Tree Performance Metrics") %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover", "conden
sed"))
```

Decision Tree Performance Metrics

Metric	Value
Accuracy	0.970
Recall (Sensitivity)	0.916
Precision	0.961

3.2 KNN

```
#Run KNN
knn_pred <- knn(train = hr_train,    # training data
                test = hr_test,      # testing data
                cl = hr_train_y,     # training labels (left column)
                k = 21)              # number of neighbors

#Check prediction output
summary(knn_pred)
```

```
##    0    1
## 3439 1060
```

4 Step 6: Evaluate the Model

```
# Generate the confusion matrix
confusionMatrix(as.factor(knn_pred),
                as.factor(hr_test_y),
                positive = "1")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##          0 3235  204
##          1  170  890
##
##                Accuracy : 0.9169
##                  95% CI : (0.9084, 0.9248)
##     No Information Rate : 0.7568
##     P-Value [Acc > NIR] : < 2e-16
##
##                   Kappa : 0.7717
##
##  Mcnemar's Test P-Value : 0.08794
##
##             Sensitivity : 0.8135
##             Specificity : 0.9501
##          Pos Pred Value : 0.8396
##          Neg Pred Value : 0.9407
##              Prevalence : 0.2432
##          Detection Rate : 0.1978
##    Detection Prevalence : 0.2356
##       Balanced Accuracy : 0.8818
##
##        'Positive' Class : 1
##
```

4.1 Business Impact Assumptions

To evaluate the financial impact of HR initiatives, we base our analysis on the following assumptions:

Metric	Value	Notes
Employees	14,999	

Metric	Value	Notes
Attrition Rate	24%	~3,600 employees/year
Cost per Employee	$20,000	Recruiting, onboarding, lost productivity
Sensitivity	81%	Correctly identifies employees who leave
Precision	83.8%	% of predicted leavers that are correct

We compare two potential HR initiatives:

Initiative	Cost per Employee	Retention Success Rate
Low-Cost Recognition Program	$500	25%
High-Impact Career Development	$1,000	35%

These assumptions allow us to estimate: - Number of leavers retained - Total cost and ROI of interventions - Savings from preventing attrition - Waste/spend on false positives - Overall effectiveness of each initiative

5 Finanaical Impact

Calculate the number of employees who leave, how many are identified by the model, and how many receive interventions.

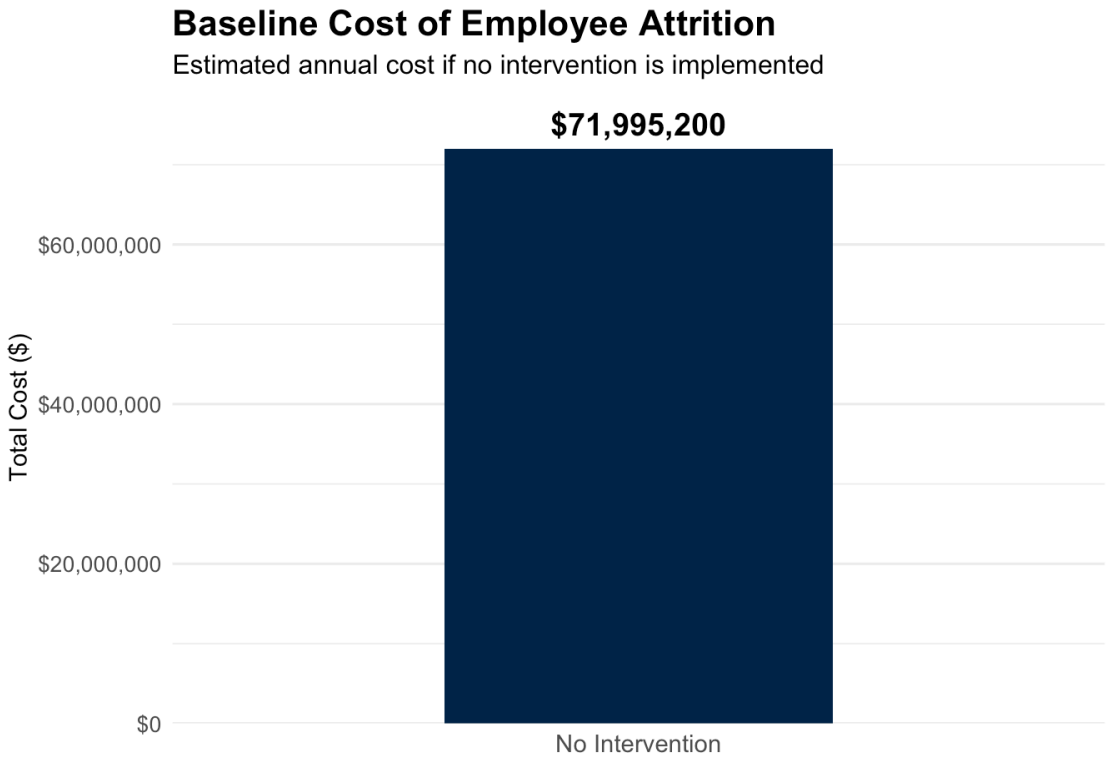
For each initiative, compute: - employees retained
- total intervention cost
- savings from retention
- net savings

```
# Assumptions
total_employees <- 14999
attrition_rate <- 0.24
turnover_cost <- 20000
```

```
#baseline costs for no interventions
total_leavers <- total_employees * attrition_rate
baseline_cost <- total_leavers * turnover_cost
baseline_cost
```

```
## [1] 71995200
```

```
ggplot(data.frame(Scenario = "No Intervention", Cost = baseline_cost),
       aes(x = Scenario, y = Cost)) +
  geom_bar(stat = "identity", fill = "#00274C", width = 0.5) +

  geom_text(
    aes(label = scales::dollar(Cost)),
    vjust = -0.6,
    size = 5,
    fontface = "bold"
  ) +

  labs(
    title = "Baseline Cost of Employee Attrition",
    subtitle = "Estimated annual cost if no intervention is implemented",
    x = "",
    y = "Total Cost ($)"
  ) +

  scale_y_continuous(
    labels = scales::dollar_format(),
    expand = expansion(mult = c(0, 0.1))  # adds space above bar
  ) +

  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    axis.text.x = element_text(size = 11),
    axis.text.y = element_text(size = 10),
    panel.grid.major.x = element_blank()
  )
```


Baseline Cost of Employee Attrition
Estimated annual cost if no intervention is implemented
$71,995,200
$60,000,000
$40,000,000
$20,000,000
$0
Total Cost ($)
No Intervention

```
# Decision tree model performance
sensitivity <- 0.9159
precision <- 0.9607

# Expected leavers per year
leavers_per_year <- total_employees * attrition_rate

# Employees correctly identified by the model
identified_leavers <- leavers_per_year * sensitivity

# Total employees flagged for intervention
adjusted_intervention_count <- identified_leavers / precision

# Recognition Program assumptions
recog_cost <- 500
recog_success <- 0.15

# Career Development Program assumptions
career_cost <- 1000
career_success <- 0.25

# Recognition Program calculations
recog_retained <- identified_leavers * recog_success
recog_savings <- recog_retained * turnover_cost
recog_total_cost <- adjusted_intervention_count * recog_cost
recog_net_savings <- recog_savings - recog_total_cost
recog_roi <- (recog_net_savings / recog_total_cost) * 100

# Career Development Program calculations
career_retained <- identified_leavers * career_success
career_savings <- career_retained * turnover_cost
career_total_cost <- adjusted_intervention_count * career_cost
career_net_savings <- career_savings - career_total_cost
career_roi <- (career_net_savings / career_total_cost) * 100

# Create results table
results <- data.frame(
  Initiative = c("Recognition Program", "Career Development Program"),
  Retained_Employees = round(c(recog_retained, career_retained)),
  Adjusted_Intervention_Count = round(c(adjusted_intervention_count, adjusted_interv
ention_count)),
  Cost_per_Intervention = c(recog_cost, career_cost),
  Total_Intervention_Cost = round(c(recog_total_cost, career_total_cost), 2),
  Savings_from_Retention = round(c(recog_savings, career_savings), 2),
  Net_Savings = round(c(recog_net_savings, career_net_savings), 2),
```

```
  ROI = round(c(recog_roi, career_roi), 1)
)

results
```

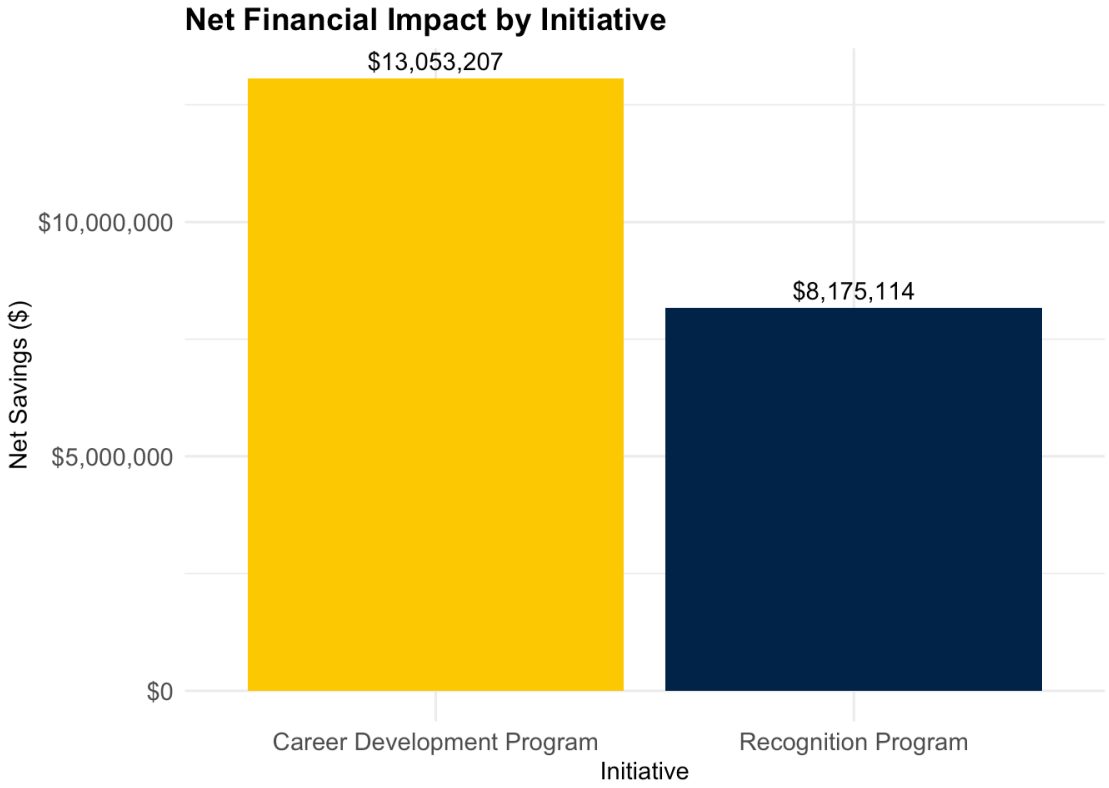
```
##                  Initiative Retained_Employees Adjusted_Intervention_Count
## 1        Recognition Program               495                       3432
## 2 Career Development Program               824                       3432
##   Cost_per_Intervention Total_Intervention_Cost Savings_from_Retention
## 1                   500                 1715947                9891061
## 2                  1000                 3431894               16485101
##   Net_Savings   ROI
## 1     8175114 476.4
## 2    13053207 380.4
```

```
results[, c("Initiative", "Net_Savings", "ROI")]
```

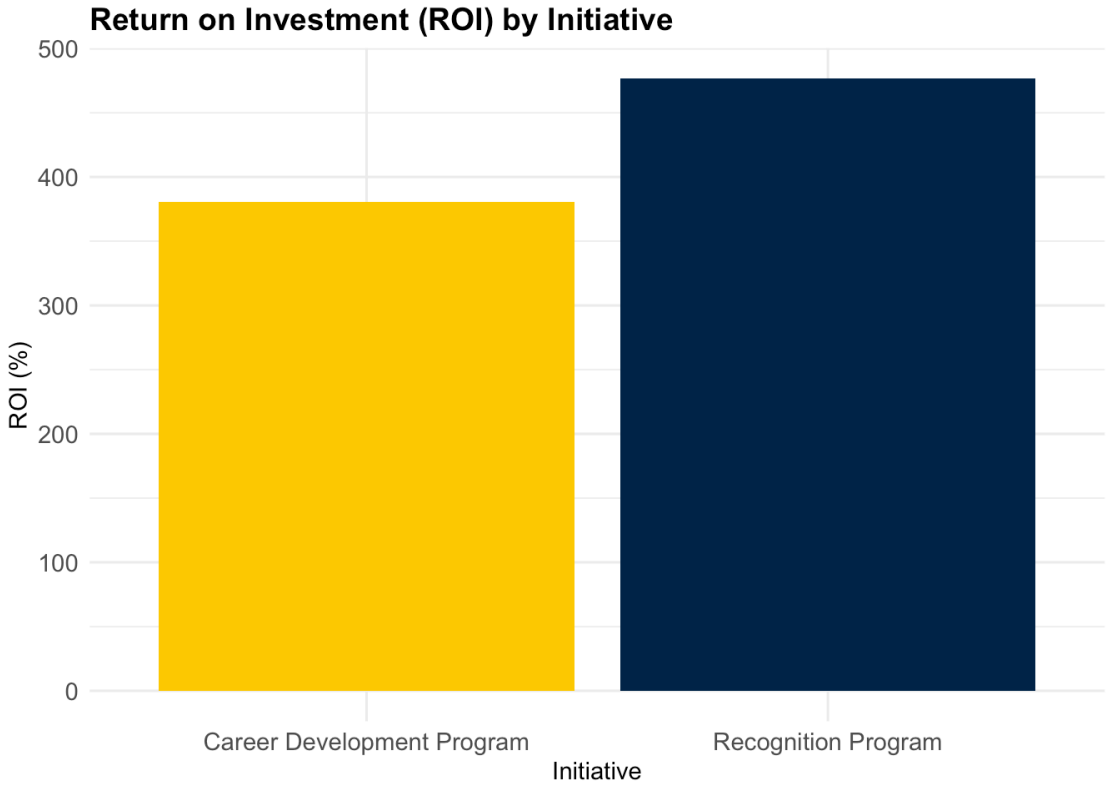
```
##                  Initiative Net_Savings   ROI
## 1        Recognition Program     8175114 476.4
## 2 Career Development Program    13053207 380.4
```

Visualize net savings for each initiative using a bar chart.

```
ggplot(results, aes(x = Initiative, y = Net_Savings, fill = Initiative)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = scales::dollar(Net_Savings)), vjust = -0.5) +
  scale_fill_manual(values = c("#FFCB05", "#00274C")) +
  labs(
    title = "Net Financial Impact by Initiative",
    x = "Initiative",
    y = "Net Savings ($)"
  ) +
  scale_y_continuous(labels = scales::dollar_format()) +
  theme_minimal() +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 14),
    axis.text = element_text(size = 11)
  )
```

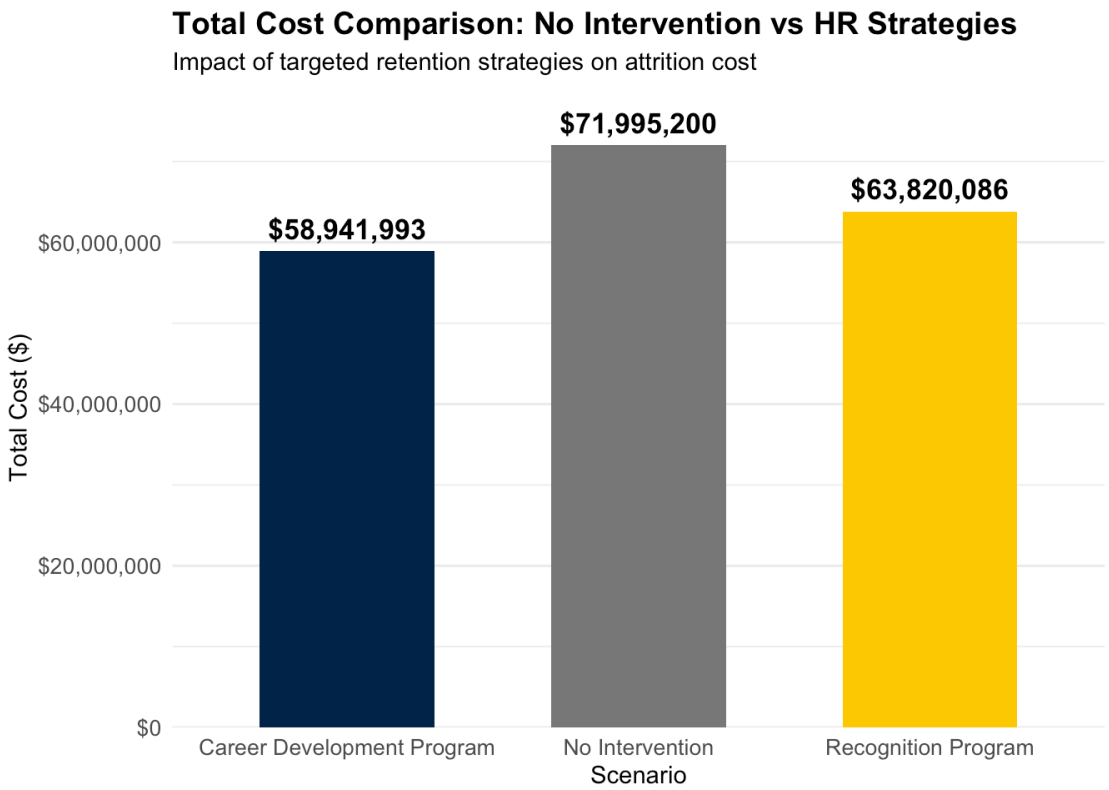

Net Financial Impact by Initiative
$13,053,207
$8,175,114
$10,000,000
$5,000,000
$0
Net Savings ($)
Career Development Program
Recognition Program
Initiative

```
results <- data.frame(
  Initiative = c("Recognition Program", "Career Development Program"),
  Retained_Employees = round(c(recog_retained, career_retained)),
  Adjusted_Intervention_Count = round(c(adjusted_intervention_count, adjusted_interv
ention_count)),
  Cost_per_Intervention = c(recog_cost, career_cost),
  Total_Intervention_Cost = round(c(recog_total_cost, career_total_cost), 2),
  Savings_from_Retention = round(c(recog_savings, career_savings), 2),
  Net_Savings = round(c(recog_net_savings, career_net_savings), 2),
  ROI = round(c(recog_roi, career_roi), 1)   # ← THIS LINE IS CRITICAL
)


ggplot(results, aes(x = Initiative, y = ROI, fill = Initiative)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("#FFCB05", "#00274C")) +
  labs(
    title = "Return on Investment (ROI) by Initiative",
    x = "Initiative",
    y = "ROI (%)"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 14),
    axis.text = element_text(size = 11)
  )
```


Return on Investment (ROI) by Initiative
500
400
300
200
100
0
ROI (%)
Career Development Program
Recognition Program
Initiative

```
# Create comparison dataset (MUST be inside this chunk or above it)
comparison <- data.frame(
  Scenario = c("No Intervention", results$Initiative),
  Cost = c(
    baseline_cost,
    baseline_cost - results$Net_Savings
  )
)

# Plot
ggplot(comparison, aes(x = Scenario, y = Cost, fill = Scenario)) +
  geom_bar(stat = "identity", width = 0.6) +

  geom_text(
    aes(label = scales::dollar(Cost)),
    vjust = -0.6,
    size = 4.5,
    fontface = "bold"
  ) +

  scale_fill_manual(values = c(
    "No Intervention" = "#7A7A7A",
    "Recognition Program" = "#FFCB05",
    "Career Development Program" = "#00274C"
  )) +

  labs(
    title = "Total Cost Comparison: No Intervention vs HR Strategies",
    subtitle = "Impact of targeted retention strategies on attrition cost",
    x = "Scenario",
    y = "Total Cost ($)"
  ) +

  scale_y_continuous(
    labels = scales::dollar_format(),
    expand = expansion(mult = c(0, 0.1))
  ) +

  theme_minimal() +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 11),
    axis.text = element_text(size = 10),
```

```
  panel.grid.major.x = element_blank()
)
```



6 Missed and Failed Interventions

Break down remaining turnover costs into two categories:

- missed leavers: employees who left but were not identified by the model
- failed interventions: employees who were identified, received an intervention, but still left

___

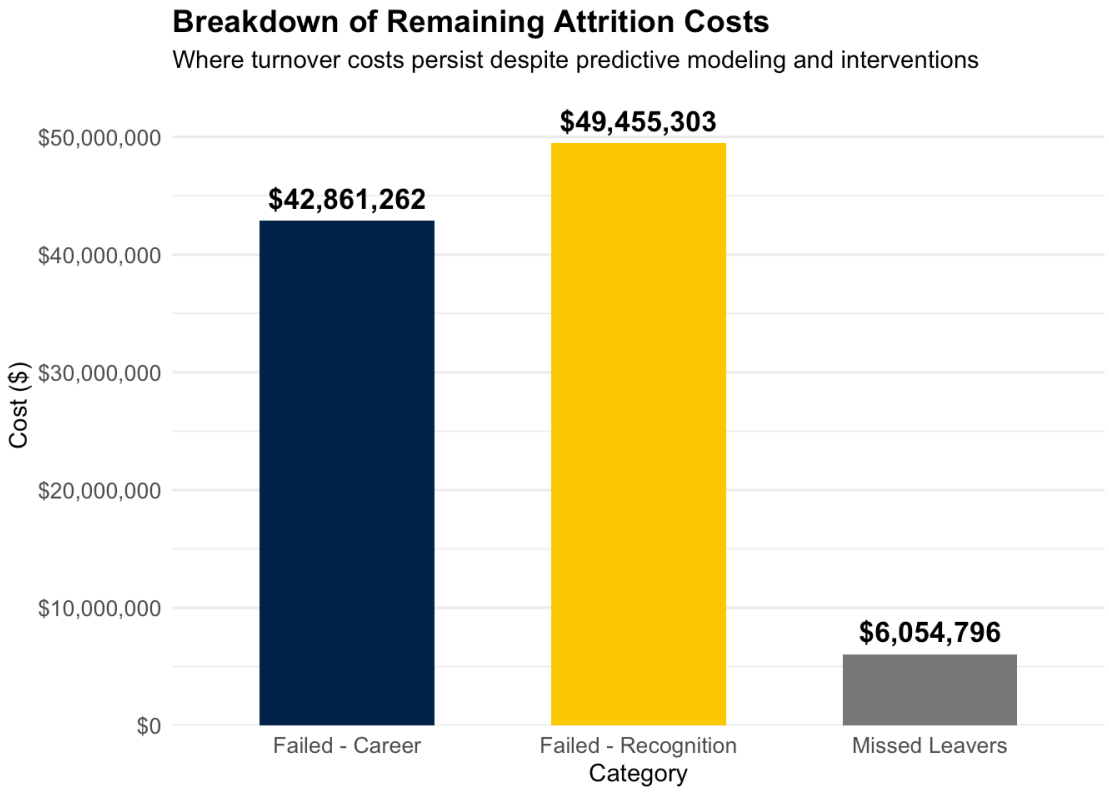
Use the attrition rate and model sensitivity to estimate total leavers, identified leavers, and missed leavers. We assume recognition program has a success rate of 25% and career success of 35%.

Then calculate how many identified employees still leave after each intervention.

```
# --- Base numbers ---
total_employees <- 14999
attrition_rate <- 0.24
turnover_cost <- 20000

# Decision Tree performance
sensitivity <- 0.9159
precision <- 0.9607

# Total leavers
total_leavers <- total_employees * attrition_rate

# Identified leavers
identified_leavers <- total_leavers * sensitivity

# Missed leavers
missed_leavers <- total_leavers - identified_leavers

# Total interventions (includes false positives)
total_flagged <- identified_leavers / precision

# --- Program assumptions ---
recog_success <- 0.25
career_success <- 0.35

# --- FAILED INTERVENTIONS ---
recog_failed <- identified_leavers * (1 - recog_success)
career_failed <- identified_leavers * (1 - career_success)

# --- COSTS ---
missed_cost <- missed_leavers * turnover_cost

recog_failed_cost <- recog_failed * turnover_cost
career_failed_cost <- career_failed * turnover_cost

# --- SUMMARY TABLE ---
failure_results <- data.frame(
  Category = c("Missed Leavers", "Failed - Recognition", "Failed - Career"),
  Employees = round(c(missed_leavers, recog_failed, career_failed)),
  Cost = round(c(missed_cost, recog_failed_cost, career_failed_cost), 2)
)

failure_results
```

```
##              Category Employees     Cost
## 1       Missed Leavers       303  6054796
## 2 Failed - Recognition      2473 49455303
## 3      Failed - Career      2143 42861262
```

```
ggplot(failure_results, aes(x = Category, y = Cost, fill = Category)) +
  geom_bar(stat = "identity", width = 0.6) +

  geom_text(
    aes(label = scales::dollar(Cost)),
    vjust = -0.6,
    size = 4.5,
    fontface = "bold"
  ) +

  scale_fill_manual(values = c(
    "Missed Leavers" = "#7A7A7A",      # gray (model failure)
    "Failed - Recognition" = "#FFCB05", # maize
    "Failed - Career" = "#00274C"       # blue
  )) +

  labs(
    title = "Breakdown of Remaining Attrition Costs",
    subtitle = "Where turnover costs persist despite predictive modeling and intervention
s",
    x = "Category",
    y = "Cost ($)"
  ) +

  scale_y_continuous(
    labels = scales::dollar_format(),
    expand = expansion(mult = c(0, 0.1))
  ) +

  theme_minimal() +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 11),
    axis.text = element_text(size = 10),
    panel.grid.major.x = element_blank()
  )
```


Breakdown of Remaining Attrition Costs
Where turnover costs persist despite predictive modeling and interventions
$50,000,000
$40,000,000
$30,000,000
$20,000,000
$10,000,000
$0
Cost ($)
$42,861,262
$49,455,303
$6,054,796
Failed - Career
Failed - Recognition
Missed Leavers
Category